

Quantum Algorithms for Approximately Evaluating Jones & HOMFLYPT Polynomials

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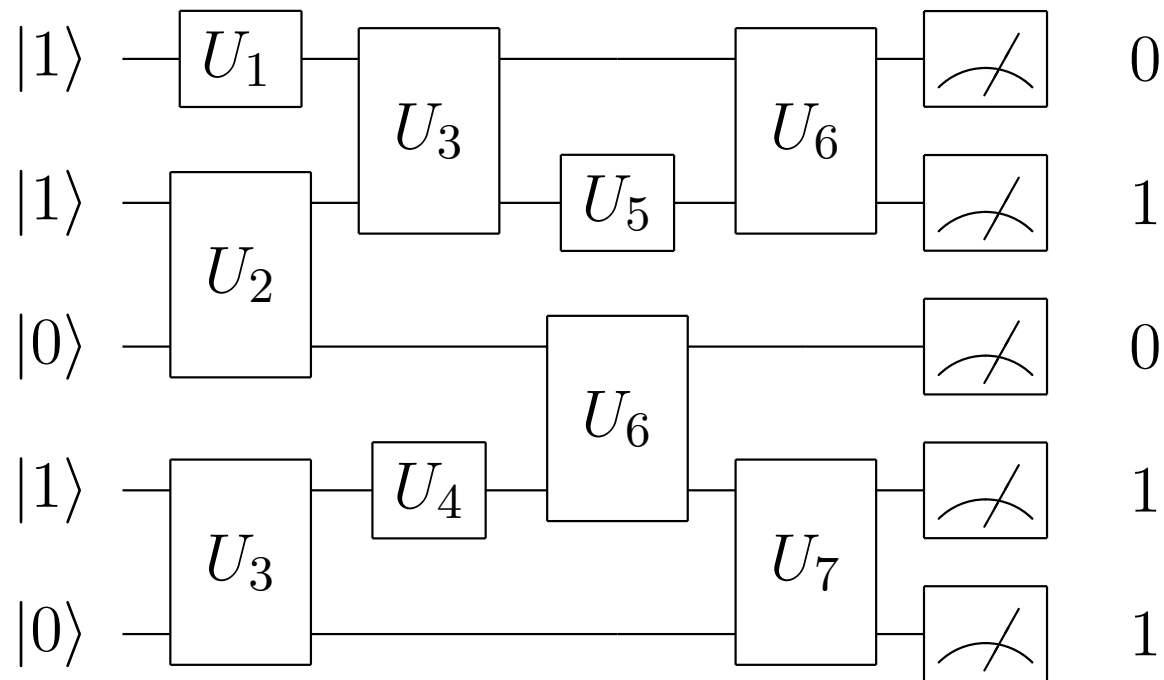
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Quantum Circuit Model

- Quantum circuits are generalizations of Boolean circuits

input transformation output (probabilistic)



Classical Bit

- Classical bit (bit): $\mathbb{B} := \{0, 1\}$
 - State: either 0 or 1
- Classical register: $\mathbb{B}^n := \underbrace{\mathbb{B} \times \mathbb{B} \times \dots \times \mathbb{B}}_n$
 - State: one of the n -bit strings

Qubit

- Quantum bit (qubit): the Hilbert space $\mathbb{C}^2$

- Basis states (classical states):

$$|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- General states: **superpositions**

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

coefficients $\alpha, \beta \in \mathbb{C}$ are called **probability amplitudes**

Quantum Register

- Quantum register:

$$(\mathbb{C}^2)^{\otimes n} := \underbrace{\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \dots \otimes \mathbb{C}^2}_n$$

2^n -dim. Hilbert space with **tensor product structure**

- General state:

$$|\psi\rangle = \sum_{x \in \mathbb{B}^n} \alpha_x |x\rangle$$

where $\sum_x |\alpha_x|^2 = 1$ and $|x\rangle = |x_1\rangle \otimes |x_2\rangle \otimes \dots \otimes |x_n\rangle$

exponentially many basis states in superposition!

Quantum Transformations

- Transformations on the quantum register $\mathcal{H} := (\mathbb{C}^2)^{\otimes n}$ are described by **unitary matrices** $U \in \mathcal{U}(\mathcal{H})$.
- Let $|\psi\rangle \in \mathcal{H}$ be the unit vector describing the state of the quantum register.

The state after the transformation U is given by

$$U |\psi\rangle$$

- Each transformations U has to be implemented by a **quantum circuit**, i.e., a sequence of **elementary gates**.

Quantum Measurement

Let

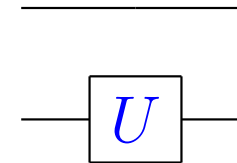
$$|\psi\rangle = \sum_{x \in \mathbb{B}^n} \alpha_x |x\rangle$$

be the state of the quantum register

- If we measure the qubits, then the probability of obtaining the outcome $x \in \mathbb{B}^n$ is given by $|\alpha_x|^2$ (the absolute value of the probability amplitude squared).

Single-Qubit Gate on Two Qubits

- single-qubit gate U on **second** qubit



- action on basis states of $\mathbb{C}^2 \otimes \mathbb{C}^2$

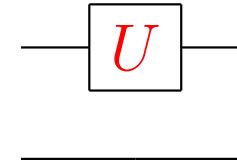
$$\begin{aligned}
 |00\rangle &\mapsto |0\rangle U|0\rangle \\
 |01\rangle &\mapsto |0\rangle U|1\rangle \\
 |10\rangle &\mapsto |1\rangle U|0\rangle \\
 |11\rangle &\mapsto |1\rangle U|1\rangle
 \end{aligned}$$

- corresponding matrix

$$I \otimes U = \left(\begin{array}{c|c} 1 \cdot U & 0 \cdot U \\ \hline 0 \cdot U & 1 \cdot U \end{array} \right) = \begin{pmatrix} u_{00} & u_{01} & 0 & 0 \\ u_{10} & u_{11} & 0 & 0 \\ 0 & 0 & u_{00} & u_{01} \\ 0 & 0 & u_{10} & u_{11} \end{pmatrix}$$

Single-Qubit Gate on Two Qubits

- single-qubit gate U on **first** qubit



- action on basis states of $\mathbb{C}^2 \otimes \mathbb{C}^2$

$$|00\rangle \mapsto (U|0\rangle) |0\rangle$$

$$|01\rangle \mapsto (U|0\rangle) |1\rangle$$

$$|10\rangle \mapsto (U|1\rangle) |0\rangle$$

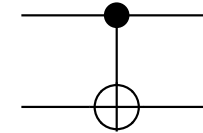
$$|11\rangle \mapsto (U|1\rangle) |1\rangle$$

- corresponding matrix

$$U \otimes I = \left(\begin{array}{c|c} u_{00} \cdot I & u_{01} \cdot I \\ \hline u_{10} \cdot I & u_{11} \cdot I \end{array} \right) = \begin{pmatrix} u_{00} & 0 & u_{01} & 0 \\ 0 & u_{00} & 0 & u_{01} \\ u_{10} & 0 & u_{11} & 0 \\ 0 & u_{10} & 0 & u_{11} \end{pmatrix}$$

Controlled-NOT Gate

- control: first qubit; target: second qubit



- action on basis states of $\mathbb{C}^2 \otimes \mathbb{C}^2$

$$\text{CNOT} |a\rangle |b\rangle = |a\rangle |a \oplus b\rangle$$

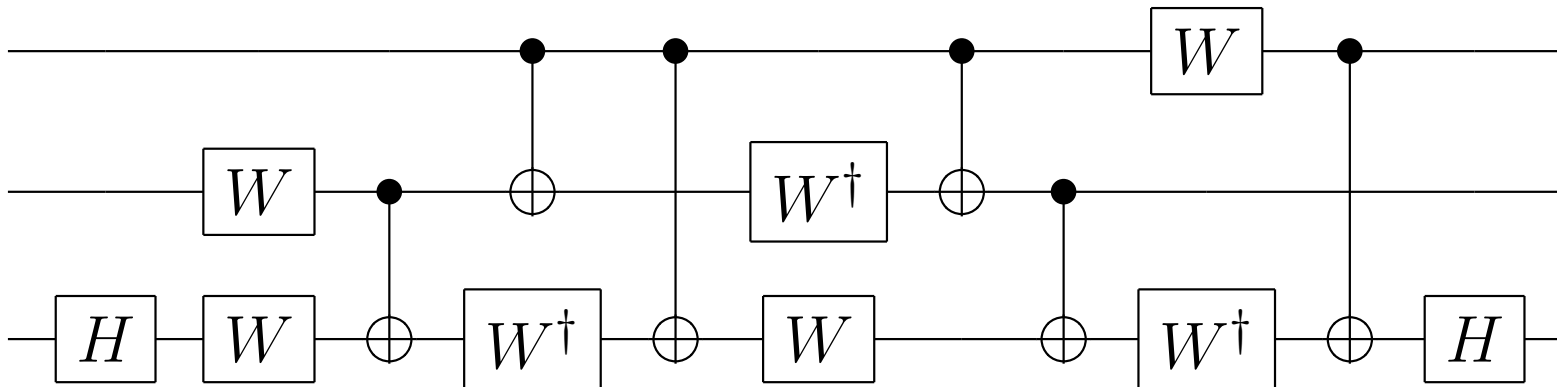
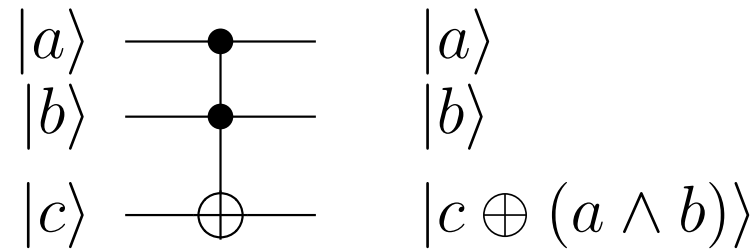
- corresponding matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Universal Gate Set

- Theorem: Each unitary $U \in \mathcal{U}(\mathcal{H})$ can be implemented by quantum circuits using only:
 - CNOT gates (acting on adjacent qubits)
 - arbitrary single qubit gates

Decomposing the Tofolli gate



$$W = \text{diag}(1, e^{i\pi/4}) \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Gate Complexity of Unitaries

- The gate complexity of a unitary $U \in \mathcal{U}(\mathcal{H})$ is minimal number of elementary gates needed to implement U .
- For example:
 - Quantum Fourier Transform has complexity $O(n^2)$
 $\implies$ Shor's factorization algorithm
 - Fast-Fourier-Transform has complexity $O(n 2^n)$

Quantum Algorithms

- Simple counting arguments show that most unitaries have **exponential** gate complexity.
- A quantum algorithm consists of
 - preparing the initial state $|x\rangle$ with $x \in \mathbb{B}^n$,
 - running a quantum circuit of **polynomially** many in n gates, and
 - measuring a subset of qubits.
- These steps are repeated polynomially many times to collect enough samples and followed by classical post-processing $\implies$ solution of the problem.

Organization I

- Hadamard Test (a very simple quantum circuit)
 - The Hadamard test makes it possible to estimate diagonal entries of unitary matrices that are implemented by quantum circuits.
- Combinatorially Block Diagonal (CBD) Unitaries
 - There is a simple method for obtaining efficient quantum circuits for implementing arbitrary CBD unitaries.
 - The CBC unitaries are not required to have any tensor product structure.

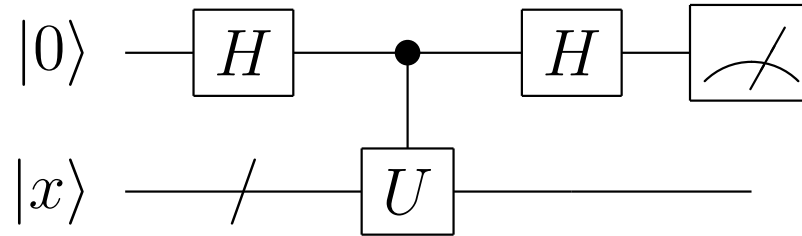
Organization II

- Jones-Wenzl Representations π_λ of the Braid Group B_n
 - The images $\pi_\lambda(\sigma_i)$ of the generators σ_i of B_n are CBD unitaries.
 - $\Rightarrow$ The images $\pi_\lambda(b)$ can be implemented efficiently for arbitrary $b \in B_n$ of polynomial length.

Organization III

- Jones and one-variable HOMFLYPT Polynomials
 - We consider the evaluation of these polynomials associated to plat and trace closures of braids b at roots of unity.
 - These evaluations can be expressed as weighted sums of diagonal entries of $\pi_\lambda(b)$.
 - We can estimate these diagonal entries with the help of the Hadamard test.

Hadamard Test



- The probabilities of obtaining the outcomes 0 and 1 are:

$$\Pr(0) = \frac{1}{2} (1 + \operatorname{Re}\langle x|U|x\rangle) \quad \Pr(1) = \frac{1}{2} (1 - \operatorname{Re}\langle x|U|x\rangle)$$

- Repeat qc $O(\epsilon^{-2} \log(\delta^{-1}))$ times $\Rightarrow$ obtain estimate R

$$|R - \operatorname{Re}\langle x|U|x\rangle| \leq \epsilon$$

with probability $1 - \delta$.

- The imaginary part can be estimated similarly.

Combinatorially Block Diagonal Unitaries

- Let V be a unitary acting on $\mathcal{H} \cong \mathbb{C}^N$.
- We call V **combinatorially block diagonal** if
 - there is some permutation P matrix such that PVP^{-1} is block diagonal,
 - the size of the largest block of PVP^{-1} is at most $d = \text{poly}(\log N)$, and
 - there is an efficient classical function that outputs the positions and values of the non-zero entries of V given the row index as input.
 - Nota bene: This makes it possible to specify EXPONENTIALLY large matrices efficiently.

Implementing CBD Unitaries

- Let $\iota : \mathcal{H} \hookrightarrow (\mathbb{C}^2)^{\otimes n}$ be some (convenient) embedding.
- There is simple method for constructing quantum circuits V such that
 - V implements U in the sense

$$\left\| V - (U_{\iota(\mathcal{H})} \oplus I_{\iota(\mathcal{H})^\perp}) \right\|_2 \leq \epsilon$$

- V consists of $\text{poly}(\log N, d, \log(1/\epsilon))$ elementary gates.

Braid group B_n

- The braid group B_n on n strands is generated by counterclockwise twists $\sigma_1, \sigma_2, \dots, \sigma_{n-1}$ which satisfy the relations:

- Yang-Baxter equation

$$\sigma_i \sigma_j \sigma_i = \sigma_j \sigma_i \sigma_j \quad |i - j| = 1$$

- far commutativity

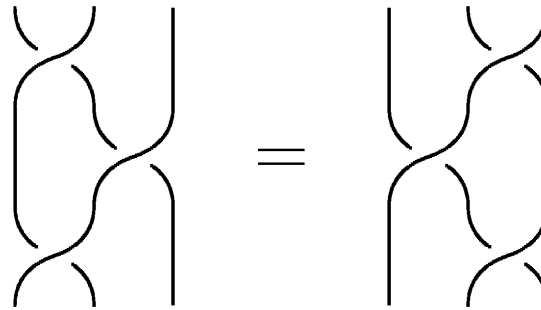
$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad |i - j| > 1.$$

Visualization

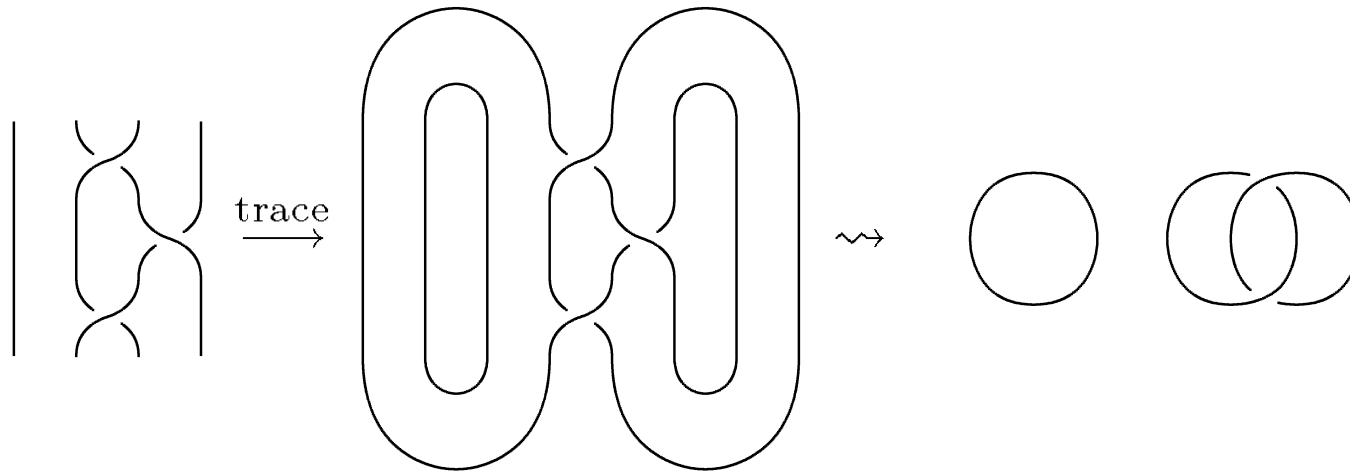
- Think of n hanging pieces of string.
- Interpret σ_i as a counterclockwise exchange of strands in positions i and $i + 1$.
- Interpret σ_i^{-1} as a clockwise exchange.
- For instance, the product $\sigma_2^{-1}\sigma_2 \in B_3$ corresponds to

$$\sigma_2^{-1}\sigma_1 = \begin{array}{c} \text{Diagram of three strands: the left and right strands cross twice, and the middle strand is straight.} \end{array}$$

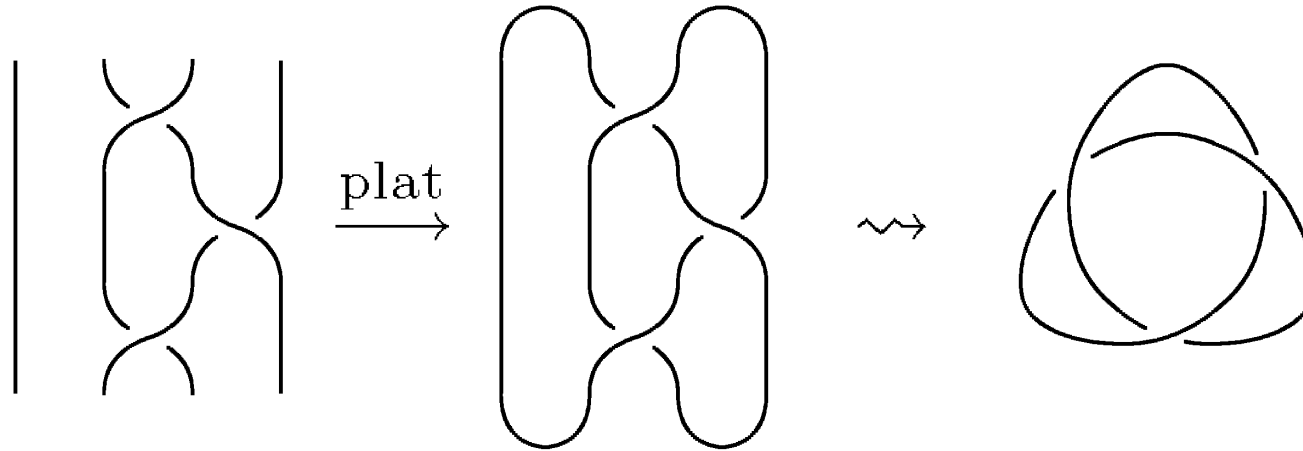
Yang-Baxter Equation



Trace closure



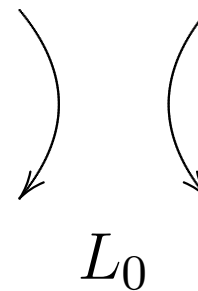
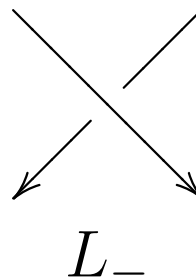
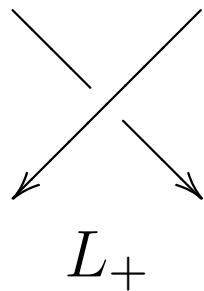
Plat closure



HOMFLYPT polynomial

- The HOMFLYPT polynomial $H_{\vec{L}}(t, x) \in \mathbb{Z}[t^{\pm 1}, x^{\pm 1}]$ is a link invariant.
- It is recursively defined by the skein relation

$$tH_{L_+}(t, x) - t^{-1}H_{L_-}(t, x) = xH_{L_0}(t, x)$$



One-Variable HOMFLYPT & Jones

- For each $k \geq 2$, we obtain the corresponding one-variable HOMFLYPT polynomial in $\mathbb{Z}[q^{\pm 1/2}]$

$$H_{\vec{L}}^{(k)}(q) = H_{\vec{L}}(q^{k/2}, q^{1/2} - q^{-1/2})$$

- Restricting to $k = 2$ yields the famous Jones polynomial

$$J_{\vec{L}}(q) \equiv H_{\vec{L}}^{(2)}(q) = H_{\vec{L}}(q, q^{1/2} - q^{-1/2})$$

Additive Approximations

- We assume that $q = e^{2\pi i/\ell}$ is an ℓ th root of unity.
- If the link $\vec{L}$ consists of m copies of the trivial knot, then

$$H_{\vec{L}}^{(k)}(q) = [k]_q^{m-1}$$

where $[k]_q$ is the quantum integer

$$[k]_q = \frac{q^{k/2} - q^{-k/2}}{q^{1/2} - q^{-1/2}} = \frac{\sin(\pi k/\ell)}{\sin(\pi/\ell)}.$$

- A value relevant for analyzing the Jones polynomial is

$$[2]_\ell = q^{1/2} + q^{-1/2} = 2 \cos(\pi/\ell).$$

Additive Approximations

- Let $b \in B_n$ be a braid of length $m = \text{poly}(n)$ and $\epsilon \in (0, 1)$ with $1/\epsilon = \text{poly}(n)$.

- Approximate HOMFLYPT Trace Closure

$$\left| H_{b^{\text{tr}}}^{(k)}(e^{2\pi i/\ell}) - Z \right| \leq \epsilon \cdot [k]_{\ell}^{n-1}$$

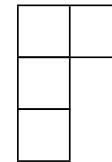
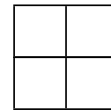
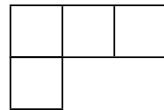
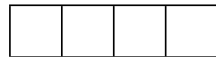
- Approximate Jones Plat Closure ($n = 2p$)

$$\left| J_{b^{\text{pl}}}(e^{2\pi i/\ell}) - Z \right| \leq \epsilon \cdot [2]_{\ell}^{p-1}$$

is the most difficult problem that can be solved efficiently by quantum computers.

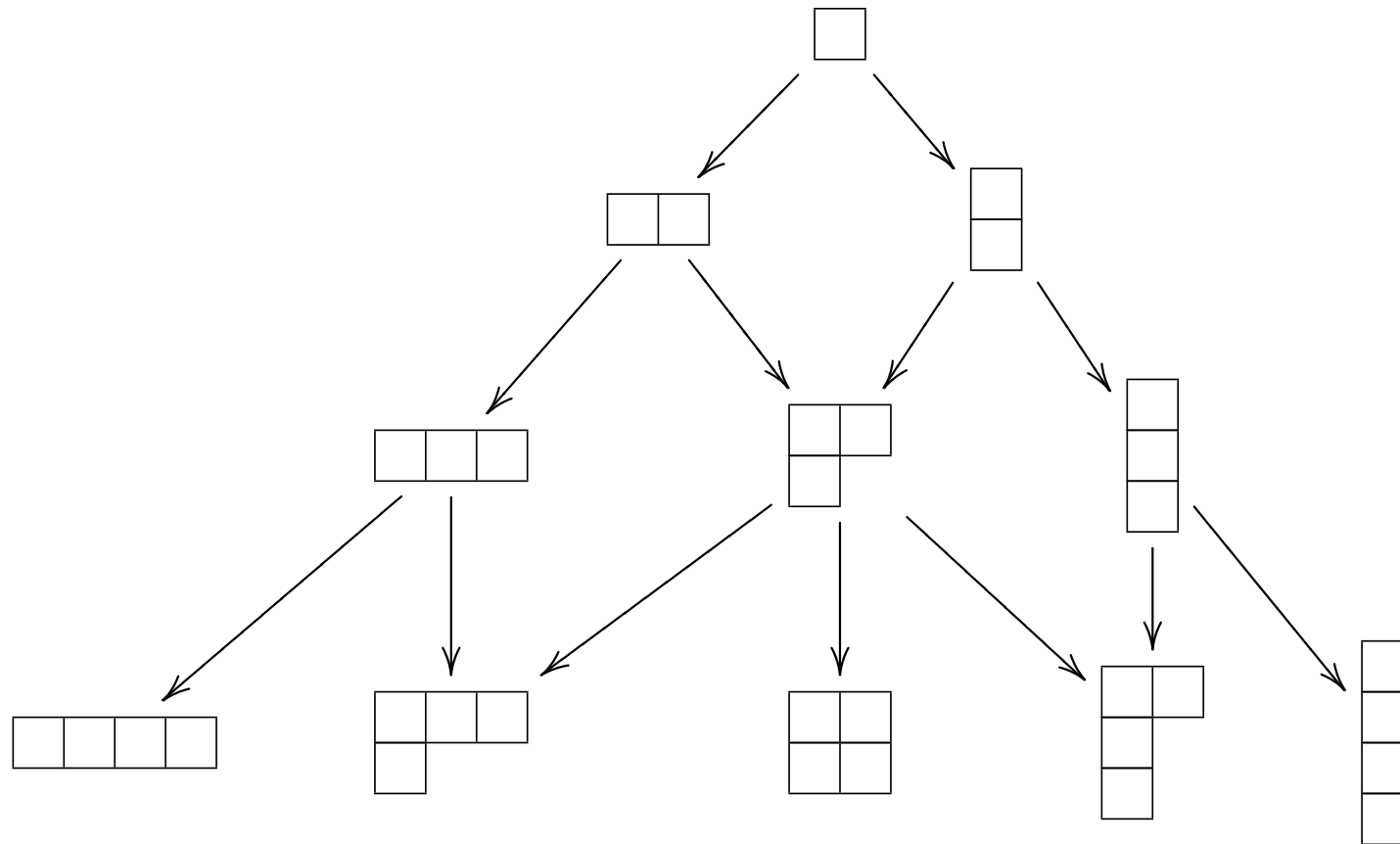
Young Diagrams

- Λ_n denotes the collection of n -box Young diagrams
- For instance, Λ_4 is given by



Young Tableaux

- T_λ denotes the collection of tableaux of shape $\lambda \in \Lambda_n$.



- The tableaux $t \in T_\lambda$ correspond bijectively to paths in the Bratteli diagram that start in $\square$ and end in λ .

(k, ℓ) -Young Diagrams

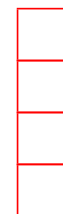
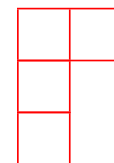
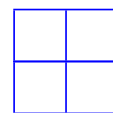
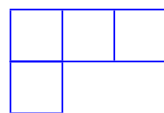
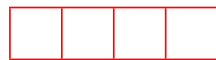
Let (k, ℓ) be a pair of integers with $\ell > k > 0$.

$\Lambda_n^{(k, \ell)}$ denotes the subcollection of (k, ℓ) -Young diagrams

$$\Lambda_n^{(k, \ell)} = \{[\lambda_1, \lambda_2, \dots, \lambda_k] \in \Lambda_n \mid \lambda_1 - \lambda_k \leq \ell - k\}.$$

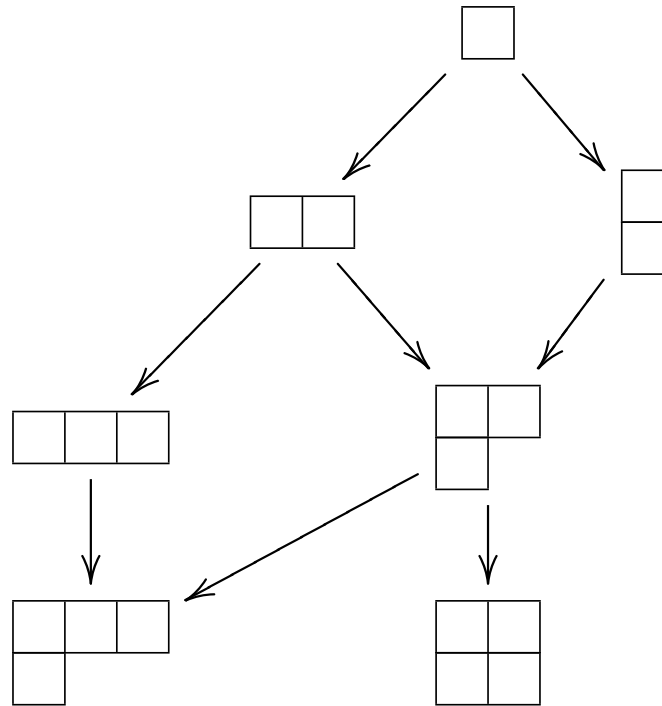
The diagrams have at most k rows and their “overhang” is at most $\ell - k$ (the parameter ℓ is called the level).

For instance, $\Lambda_4^{(2, 5)}$ is given by



(k, ℓ) -Young Tableaux

- $T_{\lambda}^{(k, \ell)}$ denotes the subcollection of (k, ℓ) -tableaux of shape $\lambda \in \Lambda_n^{(k, \ell)}$.



- The tableaux $t \in T_{\lambda}^{(k, \ell)}$ correspond bijectively to paths in the Bratteli diagram that start in $\square$ and end in λ .

Jones-Wenzl Unitary Representations

- Consider the Hilbert space

$$V_{\lambda}^{(k,\ell)} = \text{span} \left\{ |t\rangle : t \in T_{\lambda}^{(k,\ell)} \right\} .$$

- The Jones-Wenzl unitary irreducible representation

$$\pi_{\lambda} : B_n \rightarrow U \left(V_{\lambda}^{(k,\ell)} \right)$$

has the property that the images $\pi_{\lambda}(\sigma_i)$ of the generators σ_i of B_n are combinatorially block-diagonal (CBD) of size 2.

Jones-Wenzl Unitary Representations

- Define $s_i(t)$ to be either
 - the tableau that results from swapping the positions of i and $i + 1$ in t if this operation yields a valid (k, ℓ) -tableau or
 - t otherwise.

Jones-Wenzl Unitary Representations

- The images $\pi_\lambda(\sigma_i)$ are CBD of size 2 since

$$\langle t | \pi_\lambda(\sigma_i) | t' \rangle \neq 0 \quad \Rightarrow \quad t' \in \{t, s_i(t)\}.$$

- The entries are simple expressions involving k , ℓ , and the hook length between i and $i + 1$ in $t \in T_\lambda^{(k, \ell)}$.
- $\Rightarrow$ We can construct an efficient quantum circuit that implements $\pi_\lambda(b)$ for any braid b of polynomial length.

Representation-theoretic Jones Plat

- Let $\lambda = [p, p]$ be the “rectangular” two-row Young diagram ($n = 2p$).
- Let t be the tableau

$$t = \begin{array}{|c|c|c|c|c|} \hline 1 & 3 & 5 & \cdots & n-1 \\ \hline 2 & 4 & 6 & \cdots & n \\ \hline \end{array}.$$

- Jones polynomial of b^{pl} evaluated at $e^{2\pi i/\ell}$ is given by

$$J_{b^{\text{pl}}}(e^{2\pi i/\ell}) = [2]_{\ell}^{p-1} \cdot \langle t | \pi_{\lambda}(b) | t \rangle$$

Representation-theoretic HOMFLYPT Trace

- HOMFLYPT poly of b^{tr} evaluated at $e^{2\pi i/\ell}$ is given by

$$H_{b^{\text{tr}}}^{(k)}(e^{2\pi i/\ell}) = [k]_{\ell}^{n-1} \cdot \alpha \cdot \sum_{\lambda \in \Lambda_n^{(k,\ell)}} p_{\lambda}^{(k,\ell)} \cdot \frac{1}{d_{\lambda}^{(k,\ell)}} \cdot \sum_{t \in T_{\lambda}^{(k,\ell)}} \langle t | \pi_{\lambda}^{(k,\ell)}(b) | t \rangle$$

- The number $\alpha \in \mathbb{C}^{\times}$ is easy to compute from b .
- We can efficiently sample the irreps λ according to the probability distribution $p_{\lambda}^{(k,\ell)} = s_{\lambda}^{(k,\ell)} d_{\lambda}^{(k,\ell)}$

Product of the Markov weight and the dimension of $\pi_{\lambda}^{(k,\ell)}$